

The $(\infty, 1)$ -category of $(\infty, 1)$ -categories in simplicial type theory

Ulrik Buchholtz¹ Daniel Gratzer² Jonathan Weinberger³

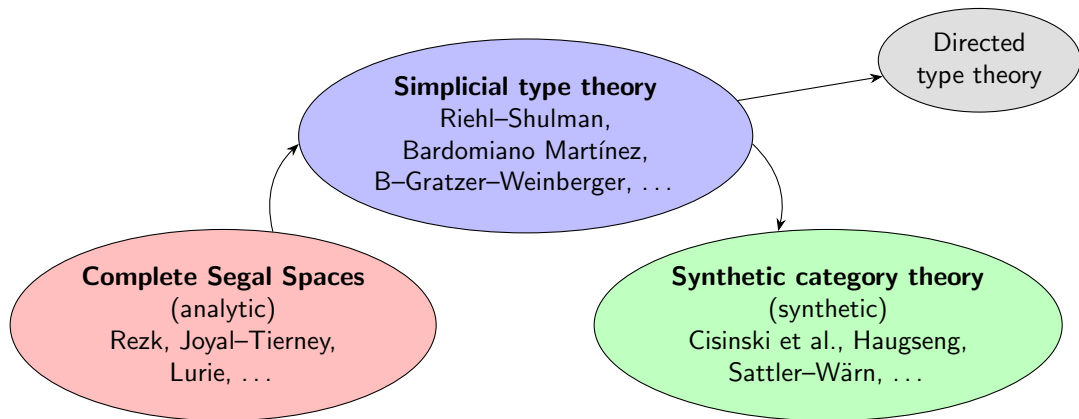
University of Nottingham
Nottingham, United Kingdom

Aarhus University
Aarhus, Denmark

Chapman University
Orange, CA, USA

HoTT/UF, Aarhus, June 2026

Simplicial Type Theory as a Bridge



Simplicial type theory functions as a bridge, working with simplicial spaces synthetically to make the analytic theory of higher categories easier.

Simplicial type theory

In simplicial type theory [RS17] we interpret HoTT in simplicial spaces (or simplicial objects of any higher topos). These are probed by the simplices Δ^n .

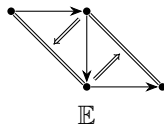
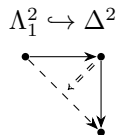
From an interval type $\mathbb{I}$ (a totally ordered set with distinct 0 and 1), we get the type of arrows $X^{\mathbb{I}}$ for any type X .

Definition X is *Segal* if $X^{\Delta^2} \rightarrow X^{\Lambda_1^2}$ is an equivalence.

Definition A Segal type X is *Rezk* if $X^{\mathbb{E}} \rightarrow X$ is an equivalence, for $\mathbb{E}$ the “walking equivalence”.

The Rezk types are categories; Segal types are precategories. It's possible to do quite a lot of abstract category theory with these, incl. fibered category theory [BW23].

Proof assistant RZK [Kud23], formalization of a version of the Yoneda lemma [KRW04].



Modal simplicial type theory

But there aren't many *examples* of categories in bare simplicial type theory. In particular, we'd like the category of spaces $\mathcal{S}$ and the category of categories, $\mathbf{Cat}$.

We define these using the [LOPS18] construction, but we need to be in a setting where $\mathbb{I}$ is *tiny*¹, which it isn't in simplicial spaces, but it is in cubical spaces: presheaves on the full subcategory on $\{(0 < 1)^n\}$ of $\mathbf{Cat}$.

With MTT [GKNB21; GKNB20], we add modalities

- idempotent comonad $\flat$ (global/objectwise) with right adjoint $\sharp$,
- involutive op (opposite),
- tw (twisted arrows) with 2-cells $\text{tw} \rightarrow \text{id}$, $\text{tw} \rightarrow \text{op}$,
- equations $\flat = \flat\flat = \flat\sharp = \flat\text{op} = \text{tw}\flat$, ...

Axioms include: $\mathbb{I}$ is a tiny bounded distributive lattice with $\langle \flat \mid 2 \rangle \simeq \langle \flat \mid \mathbb{I} \rangle$ and de Morgan duality $\langle \text{op} \mid \mathbb{I} \rangle \simeq \mathbb{I}$.

¹ $\mathbb{I}$ is tiny if $\mathbb{I} \rightarrow _$ has a right adjoint.

See also
Riley's [Ril24],
and [WL20].

Category theory in modal simplicial type theory

We can use the modalities to state and prove standard facts in category theory:

- If $C, D :_{\flat} \mathcal{U}$ are categories and $f, g :_{\flat} C \rightarrow D$, then a natural transformation $\alpha :_{\flat} \text{hom}(f, g)$ is invertible if and only if for all $c :_{\flat} C$ the map $\alpha(c) : \text{hom}(f c, g c)$ is invertible.
- Similarly, f is invertible iff f is fully faithful and essentially surjective on objects, i.e., for $\flat$ -elements.

An adjunction $C \xrightleftharpoons[g]{f} D$ is given by a family of equivalences

$$\alpha : \prod_{c:C} \prod_{d:D} \text{hom}(f c, d) \simeq \text{hom}(f, g d).$$

- If $C, D :_{\flat} \mathcal{U}$ are categories, then $f :_{\flat} C \rightarrow D$ is a left adjoint iff for all $d :_{\flat} D$ there exists $c :_{\flat} C$ and $\epsilon :_{\flat} \text{hom}(f(c), d)$ such that the following is an equivalence for all $c' :_{\flat} C$:
 $\epsilon_* \circ f : \text{hom}(c', c) \rightarrow \text{hom}(f(c'), d).$

Triangulated type theory

Cubical spaces contain simplicial spaces as a subtopos [SW21], which we can define as the lex modality $\Box$, nullification at the family of propositions:

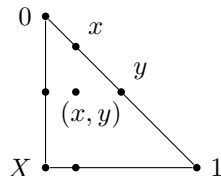
$$\Phi : \mathbb{I} \times \mathbb{I} \rightarrow \mathbf{HProp}, \quad \Phi(i, j) = (i \leq j) \vee (j \leq i)$$

The $\Box$ -modal types are the simplicial spaces. We add as axioms that cubes are projective generators [MR23].

Further, we use synthetic quasicoherence [Ble23], which relies on the fact that $\mathbb{I}$, qua cubical space, is simplicial, see also [Spi16].

The interval $\mathbb{I}$ is a distributive lattice with 0, 1. Define an $\mathbb{I}$ -algebra to be a distributive lattice R with 0, 1 and a structure-preserving map $\mathbb{I} \rightarrow R$.

Axiom (Duality) For any finitely presented $\mathbb{I}$ -algebra R , the map $R \rightarrow (\mathrm{hom}_{\mathbb{I}}(R, \mathbb{I}) \rightarrow \mathbb{I})$ is an equivalence.



Cor. (Phoa's principle) We have equivalences

$$\Delta^2 \simeq \mathbb{I}[X] \simeq (\mathbb{I} \rightarrow \mathbb{I}).$$

(Amazingly) Covariant Families

Consider the notion of covariant families:

$$\text{Cov} : \mathcal{U}^X \rightarrow \text{HProp}$$

expresses that a family is right orthogonal to $\{0\} \rightarrow \mathbb{I}$.

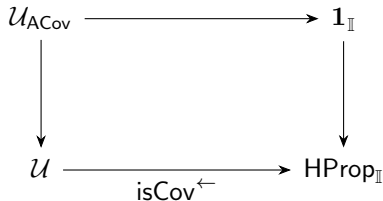
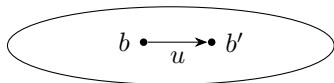
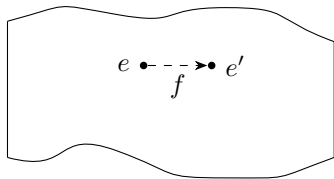
Theorem $\mathcal{S} := \mathcal{U}_{\text{ACov}} := \mathcal{U} \times_{\text{HProp}_{\mathbb{I}}} \mathbf{1}_{\mathbb{I}}$ classifies amazingly covariant families: for all $X :_{\mathcal{U}}$, the canonical map

$$\langle b \mid X \rightarrow \mathcal{S} \rangle \rightarrow \langle b \mid \sum_{A:X \rightarrow \mathcal{U}} \text{isCov}(A) \rangle$$

is an equivalence.

Closure properties $\mathcal{S}$:

- contains global discrete types,
- is closed under Σ and identity types,
- is closed under finite limits and colimits.



Directed Univalent Universe of Covariant Families

To prove directed univalence we define directed glue types, given a type X and a family $A : X \rightarrow \mathcal{S}$:

$\text{Gl } A : X \times \mathbb{I} \rightarrow \mathcal{U}_{\Box}$

$\text{Gl } A x i = (i = 0) \rightarrow A x$

Theorem $\text{Gl } A :_g X \times \mathbb{I} \rightarrow \mathcal{S}$.

This uses Phoa's principle.

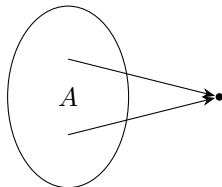
Given $E_0, E_1 : \mathcal{S}$ with $f : E_0 \rightarrow E_1$ we get a type family

$$\sum_{e_1 : E_1} \text{Gl } f^{-1}(e_1) : \mathbb{I} \rightarrow \mathcal{S}$$

This gives us straightening.

Theorem (Directed univalence)

$S : (E_0 \rightarrow E_1) \rightarrow \text{hom}_{\mathcal{S}}(E_0, E_1)$ is an equivalence.



Corollary $\mathcal{S}$ is Segal and Rezk, hence a category.

Theorem $\mathcal{S}$ is simplicial.

Currently, for this we axiomatize that $\Box A$ has the same n -simplices as A .

Directed structure identity principle

As a consequence of directed univalence we can construct many other categories with the expected morphisms:

$(\infty, 1)$ -category of pointed spaces, $\mathcal{S}_\star = \sum_{X:\mathcal{S}} X$,

1-category of finite sets,

1-categories of usual algebraic structures,

1-category of posets, ω , Δ ,

$(\infty, 1)$ -category of spectra $\mathrm{Sp} = \varprojlim (\mathcal{S}_\star \leftarrow \mathcal{S}_\star \leftarrow \dots)$,

$(\infty, 1)$ -category of spaces with an endomorphism,

...

We can prove Sp is cocomplete and stable.

(Amazingly) Cocartesian Families

To tackle the category of categories $\mathbf{Cat}$, we need to classify cocartesian families. An arrow $f : \mathbb{I} \rightarrow \tilde{A}$ in a family $A : X \rightarrow \mathcal{U}$ is *cocartesian* if diagrams of the following shape have a unique lift whenever $\Delta^{\{\bar{0} \rightarrow \bar{1}\}} \rightarrow \Delta^2 \rightarrow \tilde{A}$ is f :

$$\begin{array}{ccc} \Lambda_0^2 & \longrightarrow & \tilde{A} \\ \downarrow & & \downarrow \\ \Delta^2 & \longrightarrow & X \end{array}$$

A family $A : X \rightarrow \mathcal{U}$ is *cocartesian* if it is iso-inner, each $A(x)$ is simplicial, and the following holds:

$$\prod_{u:\mathbb{I} \rightarrow X} \prod_{a:A(u\,0)} \sum_{f:\mathrm{hom}_u^A(a,\bullet)} \mathrm{isCocartArr}_A(f)$$

In particular, in a cocartesian family, every arrow factors uniquely as a cocartesian arrow followed by a vertical arrow. A cocartesian family has functorial cocartesian transport maps.

Thanks to an observation by Rob Schellingerhout, naively locally cocartesian families, i.e., those whose restriction $A \circ u$ for any $u : \mathbb{I} \rightarrow X$ is cocartesian, are *already* cocartesian. I.e., given (u, a) , the category of lifts of u starting at a has an initial object.

(To recover the external notion of locally cocartesian family, we need to restrict to $u :_{\flat} \mathbb{I} \rightarrow X$.)

(Amazingly) Cocartesian Families

Theorem $\text{Cat} :=$

$\sum_{A:\mathcal{U}} \text{isRezk } A \times \text{isInner } A \times \text{aHasNLCCLifts } A$ is the base of the universal cocartesian family, i.e., for all $X : \mathcal{U}$, the canonical map

$$\langle \flat \mid X \rightarrow \text{Cat} \rangle \rightarrow \langle \flat \mid \sum_{A:X \rightarrow \mathcal{U}} \text{isCocart}(A) \rangle$$

is an equivalence.

Directed univalence If A, B are $\flat$ -elements of Cat , then there is a equivalence $\text{dua} : \text{hom}_{\text{Cat}}(A, B) \simeq \langle \flat \mid A \rightarrow B \rangle$.

In general, we get an equivalence between

$$\langle \flat \mid X \times \mathbb{I} \rightarrow \text{Cat} \rangle \text{ and } \langle \flat \mid \sum_{A_0, A_1 : \text{Cat}^X} A_0 \rightarrow^{\text{cc}} A_1 \rangle.$$

With a bit more work, we characterize cocartesian families $A : \flat X \times \Delta^2 \rightarrow \mathcal{U}$, allowing us to prove that Cat is Segal and Rezk.

The hardest part is then to show Cat is simplicial, which can be done by giving a suitable map back $\square \text{Cat} \rightarrow \text{Cat}$.

Examples

- The map $i : \mathcal{S} \rightarrow \text{Cat}$ is fully faithful and possesses both left and right adjoints.
- The inclusion $\text{Cat}_n \hookrightarrow \text{Cat}$ has a left adjoint.
- The category of monoidal categories MCat is the full subcategory of $\langle \text{op} \mid \Delta \rangle \rightarrow \text{Cat}$ spanned by monoidal categories, those $C^\otimes : \text{Cat}^{\langle \text{op} \mid \Delta \rangle}$ for which $(\rho_n^1, \dots, \rho_n^n) : C^\otimes([n]) \rightarrow C^\otimes([1]) \times \dots \times C^\otimes([1])$ is an equivalence for all n .
- The category of E_1 -monoids is similarly a subcategory of $\mathcal{S}^{\langle \text{op} \mid \Delta \rangle}$.

- We get a simple form of algebraic K-theory as a composite:

$$k : \text{MCat} \rightarrow \text{Mon} \rightarrow \mathcal{S}_* \xrightarrow{\Omega} \mathcal{S}_*$$

(Use that $(-)^{\simeq} : \text{Cat} \rightarrow \mathcal{S}$ is a right adjoint, inducing a functor $\text{MCat} \rightarrow \text{Mon}$.)

- $\sum_{A:\text{Cat}} A^C$ is the lax slice category.
- More in the pipeline, and much more still to do ...

Thank you

References I

- [Ble23] Ingo Blechschmidt. *A general Nullstellensatz for generalized spaces*. Draft. 2023. URL: <https://rawgit.com/iblech/internal-methods/master/paper-qcoh.pdf> (cit. on p. 6).
- [BW23] Ulrik Buchholtz and Jonathan Weinberger. “Synthetic fibered $(\infty, 1)$ -category theory”. In: *Higher Structures* 7 (1 2023), pp. 74–165. DOI: [10.21136/HS.2023.04](https://doi.org/10.21136/HS.2023.04) (cit. on p. 3).
- [GKNB21] Daniel Gratzer, G. A. Kavvos, Andreas Nuyts, and Lars Birkedal. “Multimodal Dependent Type Theory”. In: *Logical Methods in Computer Science* Volume 17, Issue 3 (July 2021). DOI: [10.46298/lmcs-17\(3:11\)2021](https://doi.org/10.46298/lmcs-17(3:11)2021) (cit. on p. 4).
- [GKNB20] Daniel Gratzer, G.A. Kavvos, Andreas Nuyts, and Lars Birkedal. “Multimodal Dependent Type Theory”. In: *Proceedings of the 35th Annual ACM/IEEE Symposium on Logic in Computer Science*. LICS '20. ACM, 2020. DOI: [10.1145/3373718.3394736](https://doi.org/10.1145/3373718.3394736) (cit. on p. 4).
- [GWB24] Daniel Gratzer, Jonathan Weinberger, and Ulrik Buchholtz. *Directed univalence in simplicial homotopy type theory*. 2024. arXiv: [2407.09146](https://arxiv.org/abs/2407.09146).

References II

- [GWB25] Daniel Gratzer, Jonathan Weinberger, and Ulrik Buchholtz. “The Yoneda embedding in simplicial type theory”. In: *40th Annual ACM/IEEE Symposium on Logic in Computer Science, LICS 2025, Singapore, June 23-26, 2025*. IEEE, 2025, pp. 127–142. DOI: [10.1109/LICS65433.2025.00017](https://doi.org/10.1109/LICS65433.2025.00017).
- [GWB26] Daniel Gratzer, Jonathan Weinberger, and Ulrik Buchholtz. *The ∞ -category of ∞ -categories in simplicial type theory*. 2026. arXiv: [2602.02218](https://arxiv.org/abs/2602.02218).
- [Kud23] Nikolai Kudasov. RZK. An experimental proof assistant based on a type theory for synthetic ∞ -categories. 2023. URL: <https://github.com/rzk-lang/rzk> (cit. on p. 3).
- [KRW04] Nikolai Kudasov, Emily Riehl, and Jonathan Weinberger. “Formalizing the ∞ -Categorical Yoneda Lemma”. In: *Proceedings of the 13th ACM SIGPLAN International Conference on Certified Programs and Proofs*. 2004, pp. 274–290. DOI: [10.1145/3636501.3636945](https://doi.org/10.1145/3636501.3636945) (cit. on p. 3).

References III

- [LOPS18] Daniel R. Licata, Ian Orton, Andrew M. Pitts, and Bas Spitters. “Internal Universes in Models of Homotopy Type Theory”. In: *3rd International Conference on Formal Structures for Computation and Deduction (FSCD 2018)*. Ed. by Hélène Kirchner. Vol. 108. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2018, 22:1–22:17. DOI: [10.4230/LIPIcs.FSCD.2018.22](https://doi.org/10.4230/LIPIcs.FSCD.2018.22) (cit. on p. 4).
- [MR23] David Jaz Myers and Mitchell Riley. *Commuting Cohesions*. 2023. arXiv: [2301.13780](https://arxiv.org/abs/2301.13780) [math.CT] (cit. on p. 6).
- [RS17] Emily Riehl and Michael Shulman. “A type theory for synthetic ∞ -categories”. In: *Higher Structures 1* (1 2017), pp. 147–224. DOI: [10.21136/HS.2017.06](https://doi.org/10.21136/HS.2017.06) (cit. on p. 3).
- [Ril24] Mitchell Riley. *A Type Theory with a Tiny Object*. 2024. arXiv: [2403.01939](https://arxiv.org/abs/2403.01939) [math.CT] (cit. on p. 4).
- [Spi16] Bas Spitters. *Cubical sets and the topological topos*. 2016. arXiv: [1610.05270](https://arxiv.org/abs/1610.05270) [cs.LO] (cit. on p. 6).

References IV

- [SW21] Thomas Streicher and Jonathan Weinberger. “Simplicial sets inside cubical sets”. In: *Theory and Application of Categories* 37.10 (2021), pp. 276–286 (cit. on p. 6).
- [WL20] Matthew Z. Weaver and Daniel R. Licata. “A Constructive Model of Directed Univalence in Bicubical Sets”. In: *Proceedings of the 35th Annual ACM/IEEE Symposium on Logic in Computer Science*. LICS '20. ACM, July 2020. DOI: [10.1145/3373718.3394794](https://doi.org/10.1145/3373718.3394794) (cit. on p. 4).